

Supplementary Results for

"A Risk-Aware Context-Adaptive Framework for PPE Safety Training: Design and Simulation Evaluation"

This supplementary document provides the complete numerical results underlying the mechanism-diagnostic, ablation, practice-monotonic falsification, and cross-domain replication analyses summarized in Sections VI-D-VI-G of the main manuscript. All results correspond to the frozen canonical implementation and the same matched simulation protocol described in the paper. These supplementary tables provide detailed computational evidence and should not be interpreted as human-participant outcome data.

TABLE S1. MECHANISM-DIAGNOSTIC RESULTS

Mean over 150 runs per policy

Method	MAE_I ↓	PRA ↑	TTA ↑	DCM ↓	NCCE ↑
Static	0.0835	0.5623	0.4936	0.3540	0.9499
Random	0.0801	0.5966	0.5227	0.3698	0.9392
Score-Adaptive	0.0754	0.6646	0.4622	0.4062	0.9499
Competence-Adaptive	0.0667	0.7244	0.4202	0.5467	0.9096
Proposed	0.0681	0.5721	0.3460	0.4672	0.8365
Oracle†	0.0859	0.4349	0.2440	0.2061	0.9127

Footnote - † Oracle is a privileged simulation-only reference. PRA and TTA compare the engine-computable risk-aware priority with its latent-reference counterpart along each policy-induced trajectory. These metrics correspond to the actual competence-targeting mechanism only for the Proposed Framework, which directly uses the risk-aware priority ranking. For the other policies, PRA and TTA characterize the diagnostic agreement of the risk-aware ranking along their induced trajectories rather than the targeting rule used by those policies. DCM and NCCE characterize policy-induced practice allocation.

TABLE S2. ABLATION RESULTS FOR THE PROPOSED FRAMEWORK

Mean over 150 runs per configuration

Configuration	CCG* ↑	RWCS* ↑	CER ↓	CMR ↓	TAR ↑	SOR ↓
Complete Proposed	0.1139	0.6099	0.8444	0.4386	0.1247	0.0890
No Contextual Risk	0.1022	0.6013	0.8511	0.4457	0.1127	0.0924
No Urgency	0.1116	0.6077	0.8467	0.4413	0.1213	0.0919
No Error History	0.1098	0.6063	0.8489	0.4432	0.1180	0.0943
No Rule Engine	0.0839	0.5808	0.9067	0.5006	0.0000	0.1990
Risk-Only	0.1457	0.6389	0.8067	0.3948	0.1733	0.2686
No Trace Persistence	0.1139	0.6099	0.8444	0.4386	0.1247	0.0890

Footnote - Bold values in the manuscript indicate the best value within the ablation configurations for each metric. Lower values are preferred for CER, CMR, and SOR. The Risk-Only configuration improves several synthetic latent and post-test outcomes but substantially increases scenario oscillation, illustrating a trade-off rather than uniform superiority. The No Trace Persistence configuration reproduces the Complete Proposed learning outcomes because trace persistence does not participate in policy selection or latent-state evolution; its contribution is limited to procedural reconstruction, auditability, and trace completeness.

TABLE S3. PRACTICE-MONOTONIC FALSIFICATION RESULTS

Mean over 150 runs per policy

Method	CCG* ↑	RWCS* ↑	CER ↓	CMR ↓	TAR ↑
Static	0.3174	0.8232	0.6489	0.1940	0.5600
Random	0.3012	0.8107	0.6622	0.2016	0.5973
Score-Adaptive	0.3174	0.8232	0.6489	0.1940	0.5600
Competence-Adaptive	0.3271	0.8335	0.6400	0.1948	0.5740
Proposed	0.3526	0.8453	0.5933	0.2022	0.5607
Oracle†	0.3538	0.8507	0.6111	0.2000	0.5400

Footnote - † Oracle is a privileged simulation-only reference. Bold values in the manuscript indicate the best deployable result for each metric. The practice-monotonic environment removes the Gaussian difficulty-competence matching term while preserving the canonical practice rate, latent decay, initial latent states, trainee profiles, policy definitions, episode budget, response-generation process, and matched stochastic pairing. This environment is used as a falsification test and is not claimed to be more realistic than the canonical challenge-matched environment.

TABLE S4. CROSS-DOMAIN PORTABILITY AND RULE-STABILITY RESULTS

Domain	Runs	Valid Runs	Rules Exercised	RAC	AR	DC	TC	CDRS	Latency (ms)
Manufacturing	150	150	4/5	0.800	1.000	1.000	1.000	1.000	0.0116
Chemical Laboratory	150	150	4/5	0.800	1.000	1.000	1.000	1.000	0.0119
Construction	150	150	3/5	0.600	1.000	1.000	1.000	1.000	0.0121
Overall	450	450	4/5	0.800	1.000	1.000	1.000	1.000	0.0119

Footnote - RAC denotes Rule-Activation Coverage. AR, DC, TC, and CDRS are software-verification indicators and are not measures of educational effectiveness. Valid Run Rate (VRR) and Scenario Validity Rate (SVR) were 1.000 in each domain. Four of the five canonical rules were exercised across the combined replication; R3 was not triggered under the evaluated trajectories. Latency is reported only as an engineering feasibility measure. These results support software-level portability and procedural stability, not equivalent pedagogical effectiveness across occupational domains.

Interpretation Scope: *The supplementary results extend the numerical detail reported in the main manuscript. They characterize computational behavior under the specified synthetic learner models, parameter settings, matched Monte Carlo protocol, and PPE knowledge bases. They do not constitute evidence of human learning effectiveness, behavioral transfer, PPE compliance improvement, or accident reduction.*